WEAKLY ASSOCIATED POINTS OF A TENSOR PRODUCT WITH A PULLED BACK SHEAF IN THE FINITE TYPE CASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the chapter's own retained minimal preamble (sources/stacks-flat/retained-preamble.tex, lines 1-85) is a verbatim excerpt of the native Xy-pic setup and of the native theorem declarations and mathematical macros, followed by the project's own reference/slogan/history display-environment definitions copied mechanically from the shared preamble; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler, and extra_packages is left empty because everything needed is already in the retained preamble. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Weakly associated points of a closed subscheme and of a quotient ring. Divisors, label divisors-lemma-weakly-associated-finite (proof line 4017); Algebra, label algebra-lemma-ass-quotient-ring (proof line 4020). Transfers the weakly associated point and the associated point of the fibre back and forth between X and the affine space in which it is embedded, which is the reduction step of the first paragraph.
- The local structure of a finite type sheaf flat at a point. Same chapter flat.tex, label proposition-finite-type-flat-at-point, invoked at proof line 4029. Produces the elementary etale neighbourhood and the open V over which the pullback of F becomes of finite presentation and flat, which is the reduction of the second paragraph. The proposition is claimed separately as S14-algebra-finite-type-flat-at-point.
- Weakly associated primes under an etale neighbourhood, and the finite presentation case. Same chapter flat.tex, labels lemma-etale-weak-assassin-up-down (proof line 4033) and lemma-weak-bourbaki-pre (proof line 4056). The first lets the statement be checked after the elementary etale neighbourhood, the second is the already proven finite presentation statement that finishes the proof. The latter is claimed separately as S14-algebra-weak-bourbaki-pre.

Source reference. Original lemma 05BY: sources/stacks-chow/tags.txt, source lines 6901–6901. Navigation pointer for \ref{algebra-lemma-ass-quotient-ring} used inside the counted span (line 4020); the argument that consumes it is retained in the span.

Date: stacks-project:flat.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 05EZ: sources/stacks-chow/tags.txt, source lines 7008–7008. Navigation pointer for \ref{divisors-lemma-weakly-associated-finite} used inside the counted span (line 4017); the argument that consumes it is retained in the span.

Source reference. Original lemma 05FP: sources/stacks-flat/pinned-original-source.tex, source lines 353–353. Navigation pointer for \ref{lemma-etale-weak-assassin-up-down} used inside the counted span (line 4033); the argument that consumes it is retained in the span.

Source reference. Original lemma 05IG: sources/stacks-flat/pinned-original-source.tex, source lines 3750–3750. Navigation pointer for \ref{lemma-weak-bourbaki-pre} used inside the counted span (line 4056); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I5: sources/stacks-flat/pinned-original-source.tex, source lines 2477–2477. Navigation pointer for \ref{proposition-finite-type-flat-at-point} used inside the counted span (line 4029); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to S$ be locally of finite type. Let $x \in X$ with image $s \in S$. Let $\mathcal{F}$ be a finite type quasi-coherent sheaf on X. Let $\mathcal{G}$ be a quasi-coherent sheaf on S. If $\mathcal{F}$ is flat at x over S, then*

$$x \in \mathit{WeakAss}_X(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}) \Leftrightarrow s \in \mathit{WeakAss}_S(\mathcal{G}) \text{ and } x \in \mathit{Ass}_{X_s}(\mathcal{F}_s).$$

Proof. In this paragraph we reduce to f being of finite presentation. The question is local on X and S, hence we may assume X and S are affine. Write $X = \mathrm{Spec}(B)$, $S = \mathrm{Spec}(A)$ and write $B = A[x_1, \ldots, x_n]/I$. In other words we obtain a closed immersion $i : X \to \mathbf{A}^n_S$ over S. Denote $t = i(x) \in \mathbf{A}^n_S$. Note that $i_*\mathcal{F}$ is a finite type quasi-coherent sheaf on $\mathbf{A}^n_S$ which is flat at t over S and note that

$$i_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}) = i_*\mathcal{F} \otimes_{\mathcal{O}_{\mathbf{A}^n_S}} p^*\mathcal{G}$$

where $p : \mathbf{A}^n_S \to S$ is the projection. Note that t is a weakly associated point of $i_*(\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G})$ if and only if x is a weakly associated point of $\mathcal{F} \otimes_{\mathcal{O}_X} f^*\mathcal{G}$, see Divisors, Lemma 05EZ. Similarly $x \in \mathrm{Ass}_{X_s}(\mathcal{F}_s)$ if and only if $t \in \mathrm{Ass}_{\mathbf{A}^n_s}((i_*\mathcal{F})_s)$ (see Algebra, Lemma 05BY). Hence it suffices to prove the lemma in case $X = \mathbf{A}^n_S$. Thus we may assume that $X \to S$ is of finite presentation.

In this paragraph we reduce to $\mathcal{F}$ being of finite presentation and flat over S. Choose an elementary étale neighbourhood $e : (S', s') \to (S, s)$ and an open $V \subset X \times_S \mathrm{Spec}(\mathcal{O}_{S',s'})$ as in Proposition 05I5. Let $x' \in X' = X \times_S S'$ be the unique point mapping to x and s'. Then it suffices to prove the statement for $X' \to S'$, x', s', $(X' \to X)^*\mathcal{F}$, and $e^*\mathcal{G}$, see Lemma 05FP. Let $v \in V$ the unique point mapping to x' and let $s' \in \mathrm{Spec}(\mathcal{O}_{S',s'})$ be the closed point. Then $\mathcal{O}_{V,v} = \mathcal{O}_{X',x'}$ and $\mathcal{O}_{\mathrm{Spec}(\mathcal{O}_{S',s'}),s'} = \mathcal{O}_{S',s'}$ and similarly for the stalks of pullbacks of $\mathcal{F}$ and $\mathcal{G}$. Also $V_{s'} \subset X'_{s'}$ is an open subscheme. Since the condition of being a weakly associated point depend only on the stalk of the sheaf, we may replace $X' \to S'$, x', s', $(X' \to X)^*\mathcal{F}$, and $e^*\mathcal{G}$ by $V \to \mathrm{Spec}(\mathcal{O}_{S',s'})$, v, s', $(V \to X)^*\mathcal{F}$, and $(\mathrm{Spec}(\mathcal{O}_{S',s'}) \to S)^*\mathcal{G}$. Thus we may assume that f is of finite presentation and $\mathcal{F}$ of finite presentation and flat over S.

Assume f is of finite presentation and $\mathcal{F}$ of finite presentation and flat over S. After shrinking X and S to affine neighbourhoods of x and s, this case is handled by Lemma 05IG. □